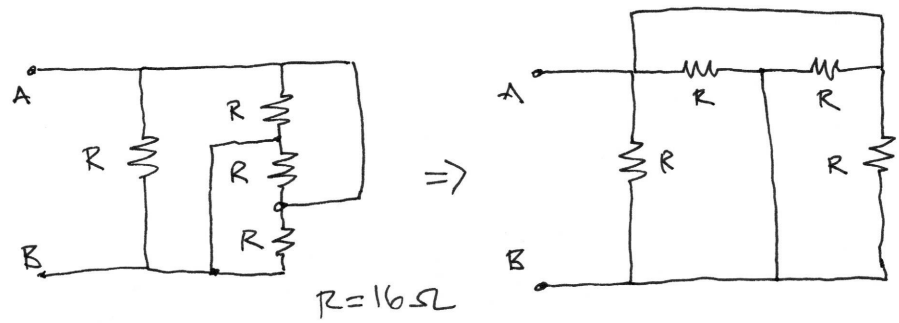
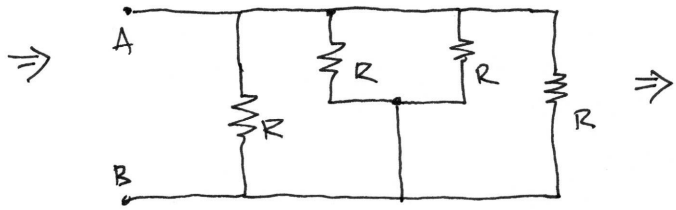
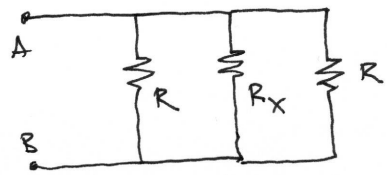
LÖSNINGSFÖRSLAG TILL

E0013E, 251218

1.







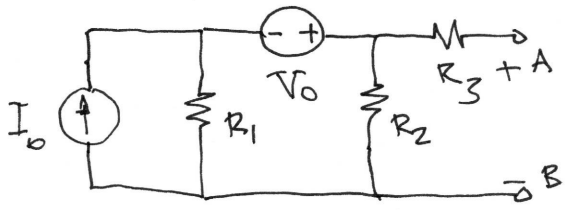
$$R_X = \frac{R^2}{2R} = \frac{R}{2} = \frac{16}{2} = 8\,\Omega$$



$$R_{AB} = \left[\frac{1}{16} + \frac{1}{8} + \frac{1}{16}\right]^{-1} = \left[\frac{1}{4}\right]^{-1} = 4\,\Omega$$

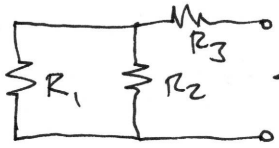
SVAR: $R_{AB} = 4\,\Omega$

2.



KORTSLUT SP. KÄLLAN OCH TA BORT STRÖMKÄLLAN

$I_0 = 3\,mA$
$V_0 = 4\,V$
$R_1 = 2k\Omega$
$R_2 = 8k\Omega$
$R_3 = 4k\Omega$

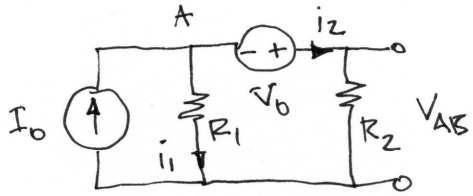


$$R_{TH} = R_3 + \frac{R_1 R_2}{R_1 + R_2} =$$

$$= 4000 + \frac{2000 \cdot 8000}{2000 + 8000} =$$

$$= \underline{5600\ \Omega}$$

ALLT TILLSAMMANS



FÖR NOD A: $I_0 - i_1 - i_2 = 0$ (1)

$$\begin{cases} i_1 = \dfrac{V_A}{R_1} \\ i_2 = \dfrac{V_A + V_0}{R_2} \end{cases} \quad (2)$$

(2) i (1) ger

$$I_0 - \frac{V_A}{R_1} - \frac{V_A + V_0}{R_2} = 0$$

$$I_0 - \frac{V_0}{R_2} = V_A \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$V_A = \frac{I_0 - \frac{V_0}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{3 \cdot 10^{-3} - \frac{4}{8000}}{\frac{1}{2000} + \frac{1}{8000}} =$$

$$= \underline{4\,V} \quad (3)$$

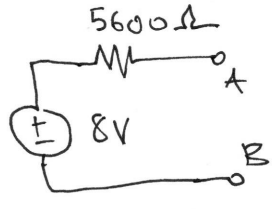
(2) & (3) ger

$$i_2 = \frac{V_A + V_0}{R_2} = \frac{4+4}{8000} = \underline{1 \cdot 10^{-3}\ A}$$

SOM GER

$$V_{AB} = R_2 i_2 = 1 \cdot 10^{-3} \cdot 8000 = \underline{8\,V} = V_{TH}$$

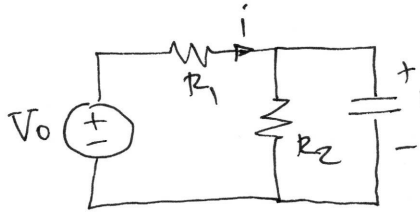
SVAR:



THEVENINEKV. BLIR

3a)

för $t \to \infty$ gäller

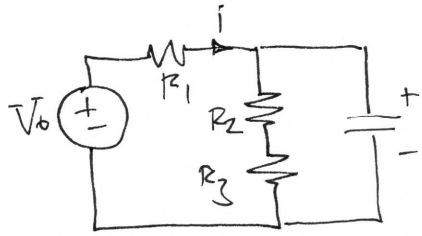


$$\Rightarrow \quad i = \frac{V_0}{R_1+R_2} = \frac{42}{48K} \simeq$$

$$\simeq 0.000875 \text{ A}$$

$$V_c(\infty) = R_2 i = 12000 \cdot 0.000875 = \underline{10.5 \text{ V}}$$

för $t \leq 0$ gäller ($t \leq 0$ ALTSÅ)



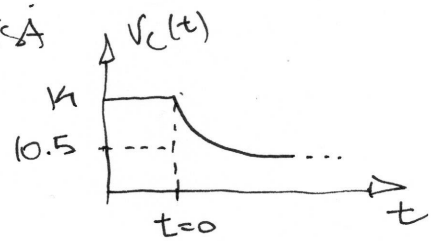
$$\Rightarrow \quad i = \frac{V_0}{R_1+R_2+R_3} =$$

$$= \frac{42}{54K} \simeq 0.00078 \text{ A}$$

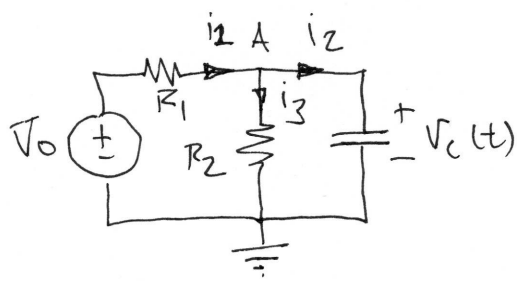
SPÄNNINGEN $V_c(0) = (R_2+R_3)\, i =$

$$= 18K \cdot 0.00078 \simeq \underline{14 \text{ V}}$$

ALTSÅ



3b) för $t \geq 0$ gäller



KSTL för NOD A ger

$$i_1 - i_2 - i_3 = 0 \qquad (1)$$

PÅ R

$$\begin{cases} i_1 = \dfrac{V_0 - V_A}{R_1} \\ i_2 = C\dfrac{dV_c}{dt} \\ i_3 = \dfrac{V_A}{R_2} \end{cases} \qquad (2)$$

MEN $V_A = V_c$ VILKET MED (1) ger

$$\frac{V_0 - V_A}{R_1} = C\frac{dV_c}{dt} + \frac{V_c}{R_2}$$

$$\frac{V_0}{R_1} = C\frac{dV_c}{dt} + \left(\frac{1}{R_1}+\frac{1}{R_2}\right)V_c$$

STUVA OM LITE

$$\frac{dV_c}{dt} + \frac{1}{C}\left(\frac{1}{R_1}+\frac{1}{R_2}\right)V_c = \frac{V_0}{R_1 C}$$

SOM HAR LÖSNING

$$V_c(t) = Ae^{-\frac{1}{C}\left(\frac{R_1+R_2}{R_1R_2}\right)t} + \frac{R_2V_0}{R_1+R_2}$$

NUMERISKT

$$V_c(t) = Ae^{-2.222t} + 10.5 \quad V.$$

MEN DETTA GÄLLER ATT $V_c(0) = 14\,V \Rightarrow$

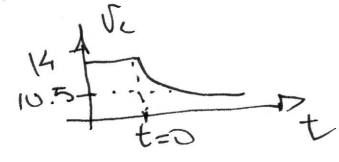
$$14 = A + 10.5 \Rightarrow A = 3.5\,V$$

ALTSÅ $V_c(t) = 10.5 + 3.5e^{-2.222t}$ VOLT

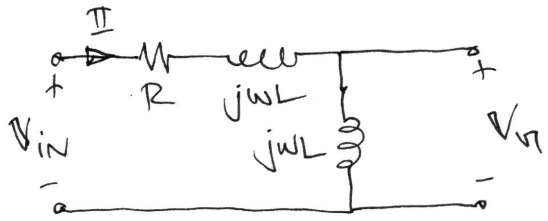
SVAR: a) $V_c(0) = 14\,V$

$V_c(\infty) = 10.5\,V$

b) $V_c(t) = 10.5 + 3.5e^{-2.222t}$ V



4.



KSPL ger

H: $V_{IN} - (R + 2j\omega L)\,I = 0$ (1)

V: $V_{UT} - j\omega L\, I = 0$

$\Rightarrow I = \dfrac{V_{UT}}{j\omega L}$ (2)

(2) i (1) ger

$$V_{IN} - (R+2j\omega L)\frac{V_{UT}}{j\omega L} = 0$$

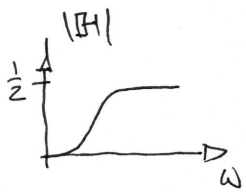
$$\frac{V_{UT}}{V_{IN}} = H(\omega) = \frac{j\omega L}{R + 2j\omega L} \qquad (1)$$

Tag Beloppet

$$|H(\omega)| = \frac{\omega L}{\sqrt{R^2 + (2\omega L)^2}}$$

$$\Rightarrow \begin{cases} |H(\omega)| = \lim\limits_{\omega \to 0} = 0 \\ |H(\omega)| = \lim\limits_{\omega \to \infty} = \dots = \frac{1}{2} \end{cases}$$

DVS.



FASFUNKTIONEN $\varphi(\omega) = \text{ATAN}\left(\frac{\text{IM}\,H}{\text{RE}\,H}\right)$ UR (1)

$$\frac{V_{UT}}{V_{IN}} = \frac{j\omega L(R - j\omega L 2)}{R^2 + (2\omega L)^2} = \frac{j\omega LR + 2(\omega L)^2}{R^2 + (2\omega L)^2} \qquad (2)$$

$$\varphi(\omega) = \text{ATAN}\left(\frac{\omega LR}{2(\omega L)^2}\right) = \text{ATAN}\left(\frac{RR}{2R^2}\right) = \text{ATAN}\left(\frac{1}{2}\right) =$$

$\omega L = R$

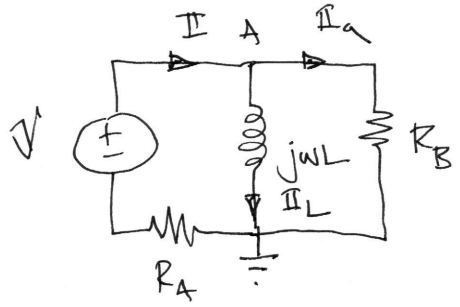
$$= 26.56^\circ$$

SVAR: a) $H(\omega) = \dfrac{j\omega L}{R + 2j\omega L}$

b) $H(\omega \to 0) = 0$, $H(\omega \to \infty) = \frac{1}{2}$

c) $\varphi\left(\omega = \frac{R}{L}\right) = 26.56^\circ$

5.)



VI VET ATT $I_L = 1\angle 0^\circ$ MA

SPÄNNINGEN I NOD A BLIR

$$V_A = j\omega L \cdot I_L = \omega L \angle 90^\circ \cdot 1 \angle 0^\circ =$$

$$= 2\pi \cdot 20 \cdot 10^3 \cdot 120 \cdot 10^{-3} \cdot 1 \cdot 10^{-3} \angle 90^\circ =$$

$$= 15.079 \angle 90^\circ \text{ VOLT.}$$

STRÖMMEN GENOM R_B BLIR

$$I_1 = \frac{V_A}{R_B} = \frac{15.079 \angle 90^\circ}{20000} \simeq 0.0007539 \angle 90^\circ \text{ A.}$$

TOTALA STRÖMMEN, KSTL I NOD A.

$$I = I_L + I_q = 1 \cdot 10^{-3} \angle 0^\circ + 0.7539 \cdot 10^{-3} \angle 90^\circ =$$

$$= 1.25 \angle 37.01^\circ \text{ MA}$$

KSPL I VÄNSTER SLINGA GER

$$V - j\omega L I_L - R_A I = 0$$

$$V = j\omega L I_L + R_A I = 15.079 \angle 90^\circ + 20 \cdot 10^3 \cdot 1.25 \cdot 10^{-3} \angle 37.0$$

$$= 19.96 + j30.13 = 36.14 \angle 56.48^\circ \text{ V}$$

SVAR: $\begin{cases} V = 36.1 \angle 56.5^\circ \text{ V} \\ I = 1.25 \angle 37.0^\circ \text{ MA} \end{cases}$